

Solutions to weekly problems and exercises: Week 5

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Week 5

1.2 What Is Arithmetic

Problem

What is the answer to a multiplication problem called? **Product**

Exercises

1. What is the answer to a multiplication problem called? **Product**
2. What is the answer to a division problem called? **Quotient**
3. What is the answer to an addition problem called? **Sum**
4. What is the answer to a subtraction problem called? **Difference**

1.3 Order of Operations: BODMAS rule

Problem

How to work this out: $10:2(2+1) = 15$

Exercises

1. $3 + 4 * 2 = 11$
2. $(1 + 2) 5 - 2 * 3 = 9$
3. $8 : 2 (2 + 1) = 12$
4. $2 - 2 + 2 = 2$
5. $(10 : 2) * 3 - 2 = 13$
6. $5 * 5^2 = 125$

1.4 Prime Numbers

Problem

Is 5 a prime number? Yes

Exercises

1. Is 13 a prime number? Yes
2. Is 33 a prime number? No

1.5 Prime Factorization

Problem

Find the prime factorization of 75. $3*5*5 = 3*5^2$

Exercises

1. Find the prime factorization of 136. $2*2*2*17 = 2^3 * 17$
2. Find the prime factorization of 100. $2*2*5*5 = 2^2 * 5^2$
3. Find the prime factorization of 124. $2*2*31 = 2^2 * 31$

1.6 Greatest Common Factor

Problem

What is the greatest common divisor of 18 and 21? 3

Exercises

1. Find the hcf of 12 and 20. 4
2. Find the gcd of 35 and 100. 5
3. Find the gcd of 3 and 12. 3

1.7 Lowest Common Factors

Problem

Find the lcm of 6 and 10. 30

Exercises

1. Find the lcm of 2 and 5. 10
2. Find the lcm of 3 and 6. 6